

llmXive Automated Review of Moebius: 0.2B Lightweight Image Inpainting Framework with 10B-Level Performance

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is advisory, automated feedback for the authors: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer_claim_accuracy

Verdict: minor_revision

The review focuses on the accuracy of factual claims and their support by the provided evidence.

****1. Statistical Significance of "Surpassing" Claims**** In Section 4.2, the authors claim Moebius "surpasses" FLUX.1-Fill-Dev on Places2 (Small) with an FID of 0.92 compared to 0.94 (Table 'cover_cmp_sota_indu'). While numerically lower, the difference of 0.02 is extremely small and likely within the standard deviation of FID measurements. The paper mentions statistical significance ($p < 0.01$) in the Supplementary Material, but the reported values there do not explicitly cover the 0.92 vs 0.94 comparison against FLUX.1-Fill-Dev. Asserting that the model "surpasses" a 10B-level industrial model based on such a marginal difference without explicit statistical validation for that specific pair is an overstatement of the evidence.

****2. Exaggeration of "Eclipsing" via Percentage Metrics**** In Section 4.2, the text states Moebius "completely eclipses" 10B-level models with "improvements of 37%–1243% in FID." The 1243% figure is derived from comparing Moebius (8.15) to SD3.5 (109.42). While mathematically correct, using such a high percentage to claim "eclipsing" is misleading. More critically, the claim ignores the teacher model, PixelHacker, which achieves 6.35 FID on FFHQ. The improvement of Moebius (8.15) over PixelHacker is actually a degradation, yet the text frames the comparison solely against the 10B models to maximize the perceived gain. The claim of "eclipsing" is not supported when the immediate predecessor is considered.

****3. Missing Citation for Commercial Comparison**** The Supplementary Material (Section ‘sup_commercial’) and Figure ‘nanobanana_qwen’ claim Moebius is “comparable” to “Nano Banana.” The citation ‘2025nano_banana’ is referenced in the text but is ****missing**** from the ‘main.bib’ file. Without a valid citation, the existence and nature of this system cannot be verified, rendering the comparative claim unsubstantiated. Additionally, the comparison is purely qualitative, lacking the quantitative metrics provided for academic baselines.

****4. Context-Dependent Speedup Claims**** The Abstract and Introduction claim a “ $>15\times$ acceleration in total inference time.” This figure is derived from Table ‘cover_cmp_sota_indu’, which compares Moebius (20 steps) against FLUX.1-Fill-Dev (50 steps). The speedup is a product of both architectural efficiency and the number of steps. The paper notes the step difference in the table caption, but the textual claim “15x faster” is presented as an inherent property. If the 10B model were run with fewer steps (e.g., 20), the speedup would drop to approximately 6x. The claim should be explicitly qualified as “15x faster under default sampling settings” to accurately reflect that the gain is partially due to the teacher’s sampling requirements.

- [writing] The claim that Moebius ‘surpasses’ FLUX.1-Fill-Dev on Places2 (Small) with FID 0.92 vs 0.94 is statistically fragile. The 0.02 difference is likely within measurement error, yet the text asserts ‘surpasses’ without citing statistical significance tests for this specific comparison.
- [writing] The claim that Moebius ‘completely eclipses’ 10B models on FFHQ with ‘improvements of 37%–1243% in FID’ is misleading. The 1243% figure compares Moebius (8.15) to SD3.5 (109.42), ignoring that the teacher PixelHacker (6.35) is a better baseline. The percentage metric exaggerates practical gains.
- [writing] The citation for ‘Nano Banana’ (2025nano_banana) in the supplementary material is missing from main.bib. The claim that Moebius is ‘comparable’ to this commercial system relies on a visual comparison without quantitative metrics and an unverifiable reference.
- [writing] The claim of ‘15x acceleration’ relies on comparing Moebius (20 steps) to FLUX (50 steps). If FLUX used 20 steps, speedup drops to ~6x. The claim should be qualified as ‘15x faster under default sampling settings’ to be accurate.

paper_reviewer_jargon_police

Verdict: full_revision

The manuscript exhibits a significant over-reliance on jargon and self-defined LaTeX macros that obscure meaning for non-specialist readers. The authors have defined nearly every technical term (e.g., ‘\gla’, ‘\glaca’, ‘\dwconv’, ‘\ffn’, ‘\kd’, ‘\lpips’, ‘\fid’, ‘\ldm’, ‘\vae’, ‘\cfg’, ‘\bfone’, ‘\muon’, ‘\sdxl’, ‘\sdthree’, ‘\crossattention’, ‘\selfattention’, ‘\gradientloss’, ‘\adaptivegradweight’) as custom LaTeX commands in the preamble. This practice forces

the reader to mentally map these cryptic abbreviations to their full meanings, creating an unnecessary barrier to entry. For instance, terms like "GLA" and "DWConv" appear frequently in the Introduction and Method sections without being spelled out in the prose itself.

Furthermore, the acronym "LCG" (Latent Categories Guidance) is used extensively throughout the paper (Abstract, Introduction, Method) but is never explicitly defined in the main text flow; it is only defined in a preamble macro. This violates the standard convention of defining acronyms at their first use. Similarly, "Mix-FFN" is introduced without a clear explanation of the "Mix" component, assuming the reader already knows the specific architectural modification.

The text also frequently uses shorthand like "10B-level" and "0.2B" without explicitly stating "billion parameters" in the initial context. While these are common in the field, a broader audience would benefit from the full phrasing at least once. The excessive use of these abbreviations and the reliance on preamble definitions rather than clear prose significantly reduces the accessibility of the paper. The authors should replace these macros with plain English text or standard abbreviations defined inline, ensuring that every technical term is clearly explained upon its first appearance.

- [writing] The manuscript overuses self-defined LaTeX macros for standard terms (e.g., `\gla`, `\glaca`, `\dwconv`, `\fn`, `\kd`, `\lpips`, `\fid`, `\ldm`, `\vae`, `\cfg`, `\bfone`, `\muon`, `\sdxl`, `\sdthree`, `\crossattention`, `\selfattention`, `\gradientloss`, `\adaptivegradweight`). These should be replaced with plain English text or standard abbreviations defined once in the text, not as custom commands, to improve readability for non-specialists.
- [writing] The acronym 'LCG' (Latent Categories Guidance) is used extensively (e.g., Abstract, Sec 1, Sec 2, Sec 3) but is not explicitly defined at its first occurrence in the main text. It is only defined in a preamble macro. Define it clearly in the first sentence where it appears.
- [writing] The term 'GLA' (Gated Linear Attention) and 'DWConv' (Depthwise Convolution) are used in the Introduction and Related Work without immediate expansion. While defined in the preamble, the text should spell them out on first use to ensure accessibility for readers unfamiliar with these specific acronyms.
- [writing] The phrase '10B-level' and '0.2B' are used repeatedly. While common in the field, the text should explicitly state '10 billion parameters' and '0.2 billion parameters' at least once in the Abstract or Introduction to avoid ambiguity for a broader audience.
- [writing] The term 'Mix-FFN' is introduced without a clear definition of what 'Mix' refers to in the context of the Feed-Forward Network. The text should briefly explain the mechanism (e.g., 'a mix of depthwise and pointwise convolutions') upon first mention.

paper_reviewer_logical_consistency

Verdict: minor_revision

The paper presents a compelling narrative for bridging the gap between lightweight and massive models, but several logical inconsistencies undermine the causal claims regarding the proposed mechanisms.

First, the motivation for the $L\lambda$ MI block relies on the premise that Gated Linear Attention (GLA) "inherently lacks the formulation to perform cross-attention" (Section 1.2). This is a logical overreach. The cited work (Yang et al., 2024) defines GLA as a mechanism that can be adapted for cross-attention by simply changing the query/key/value sources. The limitation appears to be specific to the *PixelHacker* implementation or a specific architectural choice, not a fundamental property of GLA. By framing this as a theoretical impossibility, the paper creates a strawman argument for the necessity of the Interactive- λ module. The authors must clarify whether the limitation is implementation-specific or if they are proposing a novel theoretical constraint on GLA.

Second, the proposed "Adaptive Gradient-Based Balance" (Section 3.2.2) contains a logical flaw in its stability argument. Equation 4 defines the weight $\mathcal{W}$ as the ratio of the task loss gradient norm to the distillation loss gradient norm. If the distillation loss gradient ($\|G(\mathcal{L}_{\text{F}_{\text{KD}}}, \theta_{\text{F}})\|$) becomes near-zero (a common occurrence in early training or when the student mimics the teacher well), the weight $\mathcal{W}$ approaches infinity. The text claims this mechanism provides "stability mechanisms for near-zero norms," yet the formula itself lacks any epsilon clipping or lower-bound safeguard. Without this mathematical constraint, the mechanism described is logically prone to causing gradient explosion, contradicting the claim of "rapid and smooth convergence."

Finally, the ablation study logic regarding the "capacity ceiling" is slightly circular. The paper argues that the 0.22B model hits a ceiling (FID 33.42 in Exp 10) that necessitates distillation. However, the baseline 0.526B model (Exp 1) achieves FID 32.75 *without* distillation. The logical leap that the 0.22B model's unoptimized performance is the "ceiling" is not fully supported, as the comparison is between a compressed model and a larger baseline. The argument would be stronger if the 0.22B model were compared against a 0.22B baseline with standard attention to isolate the compression effect from the optimization effect.

- [science] The claim that GLA 'inherently lacks the formulation to perform cross-attention' (Sec 1.2) contradicts the cited literature (Yang et al., 2024) which supports cross-sequence variants. Clarify if this is an implementation limit or a theoretical one.
- [science] The adaptive gradient weighting formula (Eq 4) lacks a safeguard against division by zero if the distillation gradient vanishes, contradicting the claim of 'stability mechanisms' and risking training collapse.
- [science] The ablation study (Tab. 5, Exp 10) claims the 0.22B model hits a 'ceiling' (FID 33.42) without distillation, yet the larger 0.526B baseline (Exp 1) achieves 32.75 FID

without distillation. The logic comparing different scales to define a specific model's ceiling is flawed.

paper_reviewer_overreach

Verdict: full_revision

The paper makes several aggressive claims regarding the performance of Moebius relative to 10B-parameter industrial models, which often exceed the evidence provided in the experimental results.

First, the title and abstract claim "10B-Level Performance" and that the model "rivals or even surpasses" 10B-level generalists. However, the quantitative results in Table 1 (Places2) and Table 2 (Portraits) show a mixed picture. On the Places2 Test set, Moebius (FID 9.48) is actually worse than FLUX.1-Fill-Dev (FID 8.02). While Moebius does outperform FLUX.1 on the Places2 (Small) subset, the blanket statement of "surpassing" the 10B-level SOTA is an over-generalization that ignores the specific benchmarks where the 10B model performs better. Similarly, in the portrait domain, Moebius (FID 5.39 on CelebA-HQ) underperforms its own teacher, PixelHacker (FID 4.75), and while it beats FLUX.1, the claim that it "completely eclipses" 10B models is hyperbolic given the marginal gains and the existence of a better-performing 0.86B teacher.

Second, the efficiency claim of ">15x acceleration" is derived from a comparison of total inference time using different sampling steps (20 for Moebius vs. 50 for FLUX.1). The paper does not provide evidence that the 10B models cannot achieve comparable quality with fewer steps (e.g., via distillation or scheduler optimization). Comparing a distilled student model running at 20 steps against a non-distilled teacher running at 50 steps conflates architectural efficiency with sampling efficiency, potentially overstating the architectural advantage.

Finally, the abstract states that Moebius "sets a new efficiency standard" by matching 10B performance with <2% parameters. Given that the model does not consistently match or surpass the 10B baseline across all reported metrics (specifically failing on Places2 Test FID), this claim is an over-extrapolation. The authors should temper their language to reflect that Moebius is "competitive with" or "rivals" 10B models on specific tasks, rather than claiming a universal equivalence or superiority that the data does not support. The distinction between the student's performance and the teacher's performance (PixelHacker) also needs to be clearer to avoid implying the 0.2B model has surpassed the 0.86B teacher in all aspects.

- [science] The title and abstract claim "10B-Level Performance" and that Moebius "surpasses" 10B models. However, Table 1 shows Moebius (FID 9.48) underperforms FLUX.1 (FID 8.02) on Places2 Test. The claim of surpassing is only valid for the "Places2 (Small)" subset. The title and abstract must be qualified to reflect "competitive" performance on specific benchmarks rather than a universal equivalence.
- [writing] The assertion that Moebius "surpasses" the 10B-level industrial SOTA in the

abstract is not fully supported. While it wins on Places2 (Small), it loses on Places2 Test and other metrics. The claim of "surpassing" should be restricted to the specific benchmark where data supports it, or rephrased to "rivals" to avoid over-claiming.

- [science] The claim of ">15x acceleration" compares 20 steps (Moebius) against 50 steps (FLUX.1). The paper does not justify why baselines cannot achieve similar quality with fewer steps. This conflates architectural efficiency with sampling efficiency, potentially overstating the gain if baselines were also optimized.
- [writing] The statement that Moebius "completely eclipses" 10B models in the portrait domain is an over-claim. Table 2 shows Moebius (FID 5.39) is worse than its teacher PixelHacker (FID 4.75). The term "eclipses" implies dominant superiority not supported when compared to the teacher or across all portrait metrics.

paper_reviewer_safety_ethics

Verdict: minor_revision

The manuscript includes a dedicated "Data-Privacy Impact Assessment" section, which is a positive step. However, the claim that the model employs "differential privacy-aware regularization" via weight decay and gradient clipping is technically imprecise. Standard weight decay and gradient clipping are optimization stabilizers, not formal differential privacy mechanisms (which require noise injection and specific privacy accounting). This phrasing risks misleading readers into believing the model offers formal privacy guarantees (e.g., membership inference resistance) that are not substantiated by the described methods. The authors must clarify this distinction to avoid overclaiming privacy protections.

Regarding dual-use risks, the paper highlights exceptional performance on portrait datasets (CelebA-HQ, FFHQ), achieving state-of-the-art results in facial restoration. While the "Responsible Deployment" section suggests general access controls, it lacks specific safeguards against the generation of non-consensual deepfakes or identity manipulation, a critical risk for high-fidelity portrait inpainting models. The authors should explicitly address this by recommending or implementing output watermarking, or by adding a specific clause in their ethical guidelines prohibiting the use of the model for identity reconstruction without consent.

Finally, the use of the "DeepFakeFace" dataset for Out-of-Distribution (OOD) evaluation requires careful ethical handling. Given the dataset's origin and nature, the authors should explicitly confirm in the "Data Availability" section that their usage complies with the dataset's specific ethical guidelines regarding the generation of new synthetic faces, ensuring no unintended amplification of deepfake risks occurs through their evaluation protocol.

- [writing] The manuscript includes a dedicated "Data-Privacy Impact Assessment" section, which is a positive step. However, the claim that the model employs "differential privacy-aware regularization" via weight decay and gradient clipping is technically imprecise. Standard weight decay and gradient clipping are optimization stabilizers,

not formal differential privacy mechanisms (which require noise injection and specific privacy accounting). This phrasing risks misleading readers into believing the model

paper_reviewer_scientific_evidence

Verdict: full_revision

The scientific evidence supporting the central claim—that a 0.2B model can match 10B-level performance—is currently insufficient due to gaps in statistical reporting and experimental controls.

First, the statistical significance claims in the Supplementary Materials (lines 1040-1045) are not backed by the data presented in the main text. The authors state that results are derived from “three independent runs” and report a p-value of <0.01 . However, the primary results tables (Tab. 1, Tab. 2, Tab. 3) only display single-point estimates (e.g., FID 9.48) without standard deviations or confidence intervals. Without the raw variance data or the reported standard errors in the tables, the claim of statistical significance is unverifiable and potentially misleading.

Second, the ablation study (Tab. 4) fails to isolate the contribution of the proposed architecture from the distillation strategy. The table compares the full Moebius system (Exp 9) against variants with different architectures but the same distillation setup. Crucially, it lacks a control experiment where the *original* teacher architecture (PixelHacker) or a standard 1B baseline is trained with the *same* adaptive multi-granularity distillation strategy. Without this, it is impossible to determine if the superior performance is due to the novel λ MI blocks or simply the fact that the distillation strategy is highly effective regardless of the student architecture.

Third, the comparison with industrial baselines (FLUX.1, SD3.5) is confounded by inconsistent sampling schedules. Moebius is evaluated at 20 steps, while the industrial models are evaluated at 28 or 50 steps. While the paper frames this as an efficiency advantage, it invalidates the direct quality comparison. To scientifically prove that the 0.2B model matches the 10B model’s *intrinsic* generation quality, all models must be evaluated at the same number of sampling steps (e.g., 20 steps for all). The current setup conflates architectural efficiency with the benefits of a longer sampling schedule for the larger models.

Finally, the user study (Sec. 4.3) presents preference percentages (e.g., 31.76% vs 23.70%) but provides no statistical test (e.g., binomial test) to confirm these differences are significant given the sample size (N=22 participants, 50 trials). The claim that Moebius “significantly surpasses” industrial systems based on this data is not rigorously supported.

- [science] The scientific evidence supporting the central claim—that a 0.2B model can match 10B-level performance—is currently insufficient due to gaps in statistical reporting and experimental controls. First, the statistical significance claims in the Supplementary Materials (lines 1040-1045) are not backed by the data presented in the main text. The authors state that results are derived from “three independent runs” and report a p-value of <0.01 . However, the primary results tables (Tab. 1, Tab. 2, Tab

paper_reviewer_statistical_analysis

Verdict: full_revision

The statistical analysis in the manuscript is currently insufficient to support the strong claims of superiority and equivalence, particularly regarding the comparison between the 0.2B Moebius model and 10B-level industrial baselines.

First, the claim of statistical significance in the Supplementary Materials (Section "Evaluation of Out-of-Distribution (OOD) Performance") is problematic. The authors state they computed 95% confidence intervals over "three independent runs" to achieve $p < 0.01$. However, the primary results presented in Tables 1, 2, and 3 are single-point estimates (e.g., FID 9.48, LPIPS 0.207) with no associated error bars, standard deviations, or confidence intervals. Without reporting the variance across these runs in the main tables, the reader cannot assess the reliability of the reported metrics or the validity of the significance claim. The manuscript must be revised to include standard deviation or 95% CI for all primary benchmark results.

Second, the proposed adaptive gradient-based loss weighting mechanism (Equations 4 and 5) relies on dynamic gradient norms. This introduces a stochastic element to the optimization process that is not accounted for in the experimental reporting. The paper does not provide a sensitivity analysis or a stability plot showing how these weights evolve during training, nor does it demonstrate that the final performance is robust to different random seeds. Given the extreme compression ratio, the model's performance could be highly sensitive to initialization or the specific trajectory of the adaptive weights. Reproducibility requires reporting results with fixed seeds and analyzing the variance.

Third, the user study (Section 4.3) presents preference percentages (e.g., Moebius 31.76% vs. Teacher 32.18%) but fails to report the statistical significance of these differences. With a sample size of 50 cases and 22 participants, the difference between 31.76% and 32.18% is likely not statistically significant without a formal test (e.g., binomial test or chi-squared). The authors must perform and report the appropriate statistical test to determine if the observed preference gaps are meaningful or within the margin of error.

Finally, the ablation studies (Table 4) compare different architectural components but do not include error bars or significance tests for the performance drops/gains observed. For instance, the FID shift from 25.86 to 26.43 (Exp 8 to 9) is presented as a definitive trade-off, but without variance data, it is unclear if this difference is statistically distinguishable from noise. All comparative claims in the ablation studies should be backed by statistical testing.

- [science] The statistical analysis in the manuscript is currently insufficient to support the strong claims of superiority and equivalence, particularly regarding the comparison between the 0.2B Moebius model and 10B-level industrial baselines. First, the claim of statistical significance in the Supplementary Materials (Section "Evaluation of Out-of-Distribution (OOD) Performance") is problematic. The authors state they computed 95% confidence intervals over "three independent runs" to achieve $p < 0.01$. H

paper_reviewer_writing_quality

Verdict: minor_revision

The manuscript demonstrates a high level of technical sophistication, but the writing quality occasionally suffers from minor grammatical inconsistencies and slightly awkward phrasing that impedes smooth readability. While the overall structure is logical, specific sections require polishing to ensure clarity and precision.

In Section 3.2.1, the term "spatial continuousness" is non-standard in English; "spatial continuity" is the correct and more natural phrasing. Similarly, in Section 3.2.2, the sentence structure regarding "establishing a precise spatial-semantic correspondence" is slightly clunky. Rephrasing to "Establishing a precise spatial-semantic correspondence proves challenging" would improve the flow and grammatical correctness.

In Section 4.1.2, the claim about "massive redundancy inherent in industrial foundational models" is somewhat vague. Specifying whether this refers to parameter redundancy or computational redundancy would enhance the precision of the argument. Additionally, in the Supplementary Materials under "Failure Case Analysis," the phrase "tiny background aesthetics" is ambiguous and could be misinterpreted. Replacing it with "fine-grained background details" would better convey the intended meaning regarding the model's limitations.

Overall, the paper is well-organized and the arguments are generally clear, but these minor linguistic issues should be addressed to ensure the manuscript meets the highest standards of academic writing.

- [writing] In Section 3.2.1, the phrase 'spatial continuousness' is non-standard; replace with 'spatial continuity' for grammatical correctness and clarity.
- [writing] In Section 3.2.2, the sentence '...establishing a precise spatial-semantic correspondence is challenging' lacks a clear subject-verb flow. Consider rephrasing to 'Establishing a precise spatial-semantic correspondence proves challenging' to improve readability.
- [writing] In Section 4.1.2, the phrase 'massive redundancy inherent in industrial foundational models' is slightly vague. Clarify if this refers to parameter redundancy or computational redundancy to ensure precise meaning.
- [writing] In the Supplementary Materials, Section 'Failure Case Analysis', the phrase 'tiny background aesthetics' is ambiguous. Replace with 'fine-grained background details' to better convey the intended meaning.